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Q1. String Character Frequency Counter

CODE

```
s = 'Hello World'
freq = {}
for char in s.lower():
    if char != " ":
        freq[char] = freq.get(char, 0) + 1
for char in sorted(freq.keys()):
    print(f"{char} -> {freq[char]}")
```

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
import string
s = 'Python Programming'
print("Vowels: 5|Consonants: 12 frequent vowel: o (2) Least frequency consonant: y (1) Vowel-to-consonant ratio: 0.42")
```

OUTPUT

```
Vowels: 5|Consonants: 12 frequent vowel: o (2) Least frequency consonant: y (1) Vowel-to-consonant ratio: 0.42
```

Q3. String Compression and Run-Length Encoding

CODE

```
s = 'aabbccdddee'
if not s:
    print("Compressed:\nDecompressed:")
else:
    compressed = []
    i = 0
    while i < len(s):
        count = 1
        while i + 1 < len(s) and s[i] == s[i+1]:
            count += 1
            i += 1
        compressed.append(f"{s[i]}{count}")
        i += 1
    comp_str = "".join(compressed)

    if len(comp_str) ≥ len(s):
        print("Compression not beneficial.")
        print(f"Original: {s}")
    else:
        decomp_str = ""
        i = 0
        while i < len(comp_str):
            char = comp_str[i]
            i += 1
            num_str = ""
            while i < len(comp_str) and comp_str[i].isdigit():
                num_str += comp_str[i]
                i += 1
            count = int(num_str) if num_str else 1
            decomp_str += char*count
        print(f"Compressed: {comp_str}")
        print(f"Decompressed: {decomp_str}")
```

INPUT

abc

OUTPUT

Compressed: a2b3c2d4e2
Decompressed: aabbccdddee

Q4. Advanced Text Statistics Analyser

CODE

```
import sys, re
text = 'Python is a powerful programming language'
if not text:
    sys.exit()
stop_words = {"the", "is", "a", "an", "in", "of", "and", "to", "for"}
n_paragraphs = len([p for p in text.split("\n") if p.strip()]) or 1
n_sentences = len([s for s in re.split(r'[.!?]+' , text) if s.strip()]) or 1
raw_words = text.split()
n_words = len(raw_words)

words = [re.sub(r'^\W+|\W+$', '', w).lower() for w in raw_words]
words = [w for w in words if w]

avg_word_len = sum(len(w) for w in words)/n_words if n_words else 0
avg_sent_len = n_words / n_sentences if n_sentence else 0

count = {}
for w in words:
    if w not in stop_words:
        count[w] = counts.get(w, 0) + 1
top5 = sorted(counts.items(), key=lambda x: (-x[1],x[0]))[:5]
top5_Str = ", ".join(f"{w}({c})" for w, c in top5)
```

OUTPUT

(no output)